Nature or Nurture ?
Culture or Institutions ?
Insights from Jewish History, 70 - 1850

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THE CHOSEN FEW



THE CHOSEN FEW

How Education Shaped Jewish History, 70–1492

MARISTELLA BOTTICINI and ZVI ECKSTEIN

THE CHOSEN MANY



THE CHOSEN MANY

How Culture and Institutions Shaped Jewish History, 1492–1850

MARISTELLA BOTTICINI ZVI ECKSTEIN ANAT VATURI



THE CHOSEN FEW

How Education Shaped Jewish History, 70–1492

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The Chosen Few

How Education Shaped Jewish History, 70-1492

from

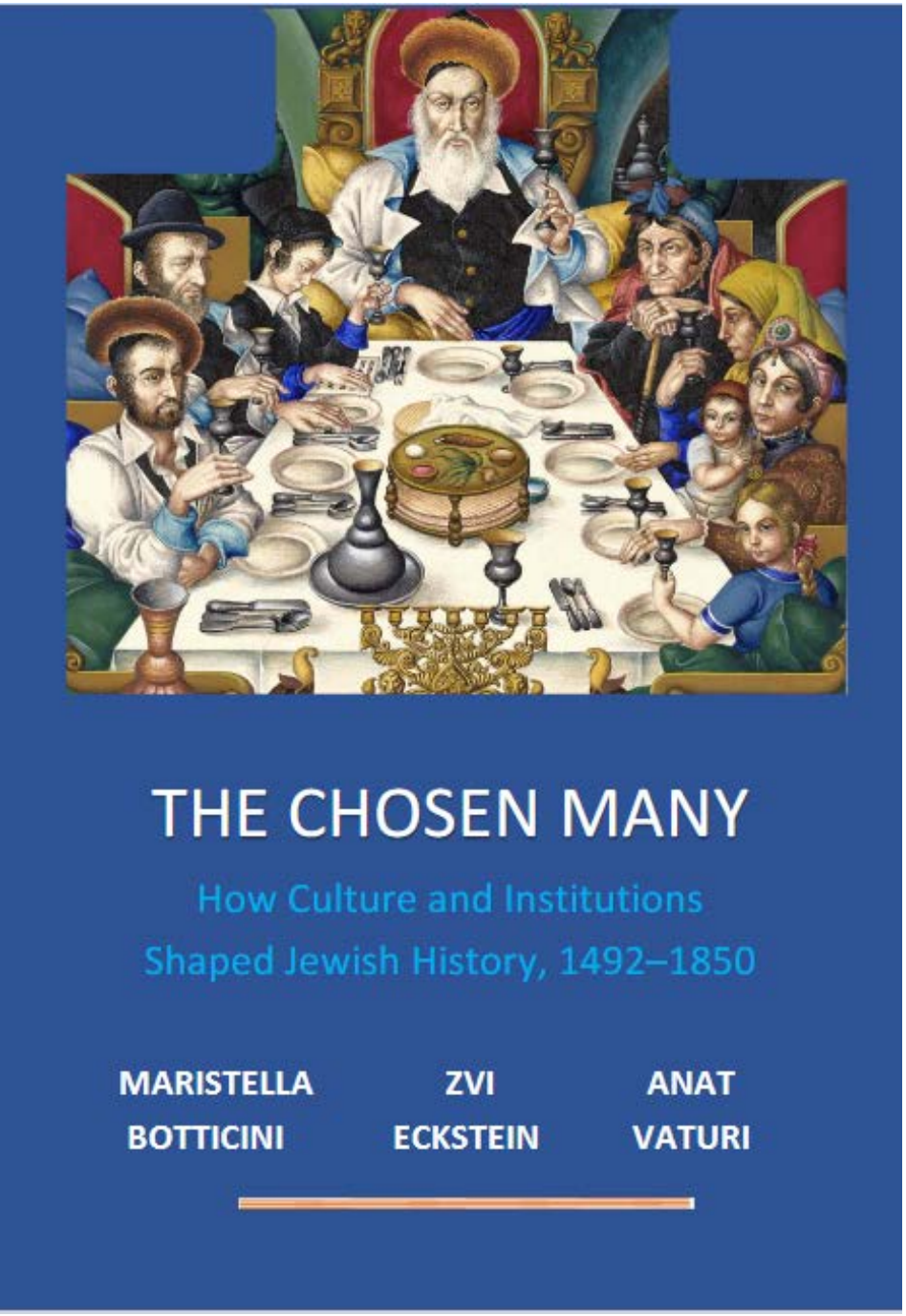
5.5 million c. 70

(illiterate farmers)

to

1 million c. 1492

(educated merchants)



The Chosen Many

How Culture and Institutions Shaped Jewish History, 1492-1850

from

1 million c. 1492

to

5 million c. 1850

specialized in skilled jobs

In a nutshell …

Q1: NATURE or NURTURE ?

Nurture

Jewish parental investment in

- infant and child care
- children education

fostered by religious / social norms

In a nutshell …

Q2: CULTURE or INSTITUTIONS ?

Both

Jewish parental investment in

- infant and child care
- children education

fostered by religious / social norms (culture)

&

- Jewish communal (internal) institutions
- economic and political institutions of Muslim Empire and Polish-Lithuanian Commonwealth



THE CHOSEN FEW

How Education Shaped Jewish History, 70–1492

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How Education Shaped Jewish History, 70-1492

An economic puzzle (from Simon Kuznets)

Country	Year	% Jews in NA jobs	% non-Jews in NA jobs	Jews as % total pop
Poland	1931	**96**	47	9.8
Lithuania	1937	**96**	---	7.6
Hungary	1930	**97**	52	5.1
Latvia	1930	**99**	47	4.8
Romania	1930	**96**	37	4.2
U.S.	1940	**98**	82	3.7
Czech	1930	**91**	73	2.4
Soviet Un.	1926	**96**	27	2.1
Argentina	1935	**94**	---	2.0
Canada	1931	**99**	71	1.5
Bulgaria	1926	**99**	31	0.9
Germany	1933	**99**	83	0.8

Why are there so few Jews in farming?

Gary Becker and George Stigler
(University of Chicago)

"De gustibus non est disputandum"
Journal of Political Economy (1977)

A historical journey through the lens of economics

To answer the question, we need to look back (very back…) in time, analyze **the history of the Jews** over 15 centuries

70 – 1492

and apply **the lens of economic theory**

[Galileo's telescope]

Ch 1 **Jewish population, locations, and occupations**

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Ch 10 1492 to today: open questions

Economic / Demographic Puzzles

- **Jewish population dynamics**

  65-650 decreased 5.5 to 1.2 million

  1250-1500 decreased 1.5 to 1 million

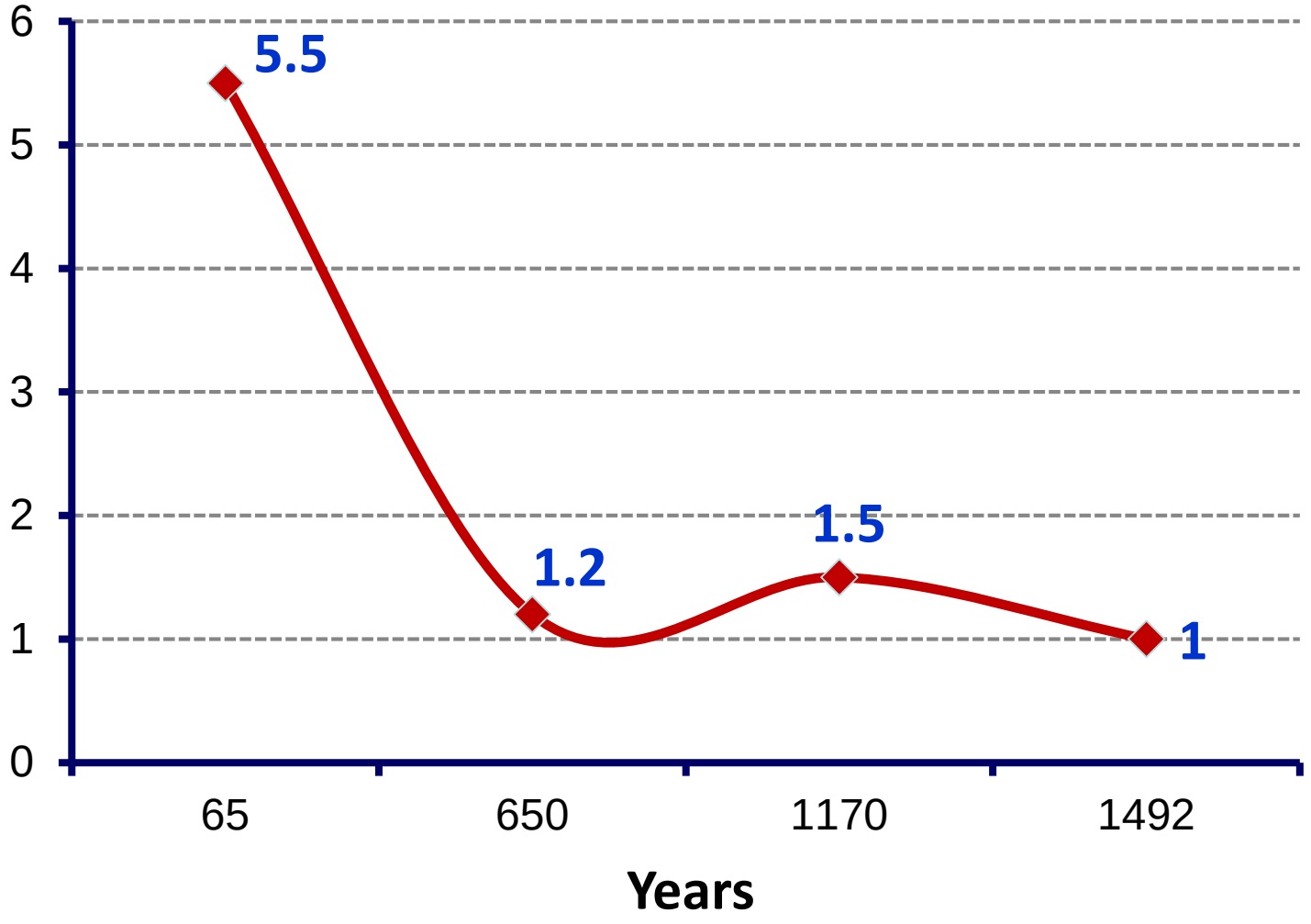
- **Occupational transition (750-900, Muslim Middle East)**

  Jews left farming and entered urban, skilled occupations

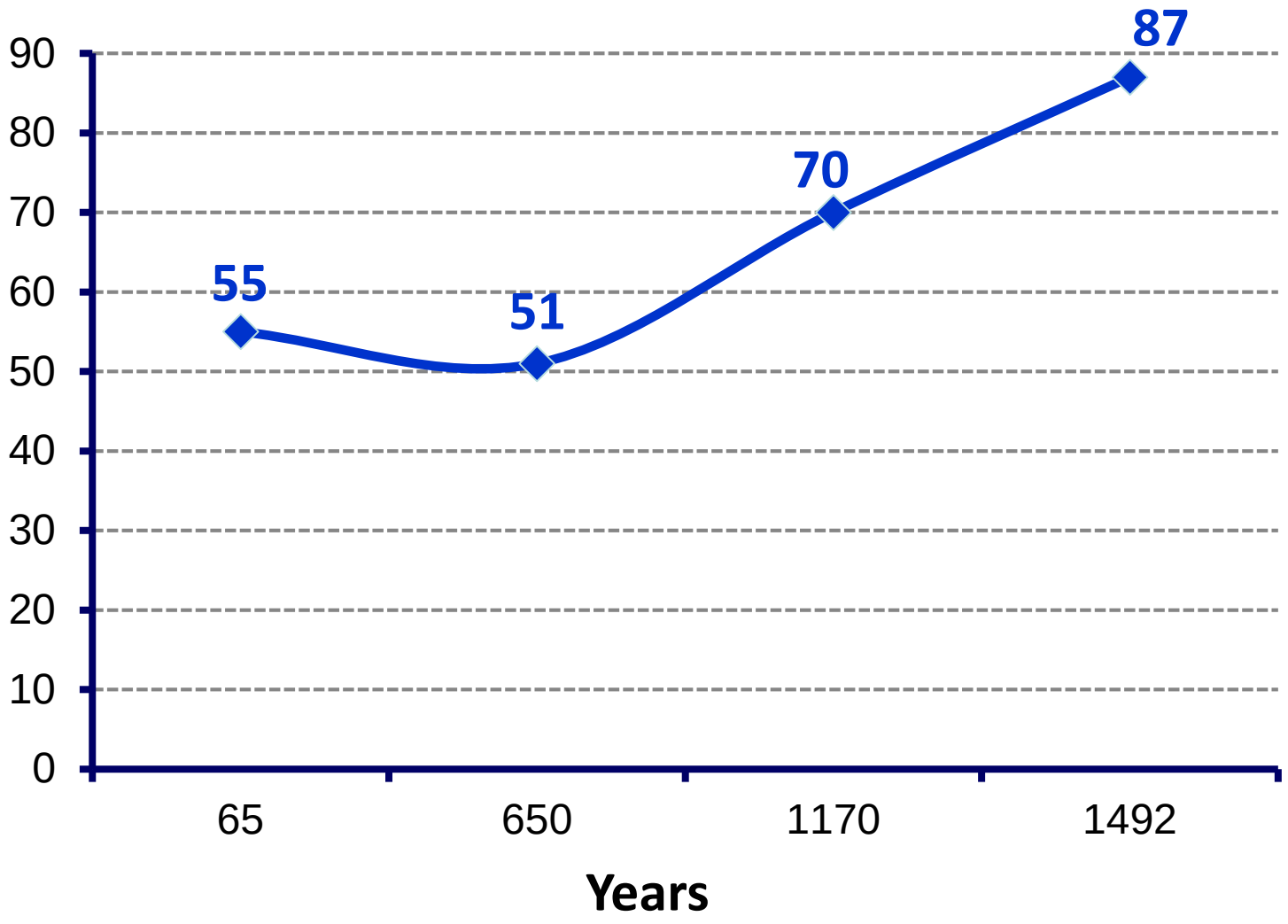
- **Jewish Diaspora and minority status (800-1200)**

  The migrations of Jewish *skills*

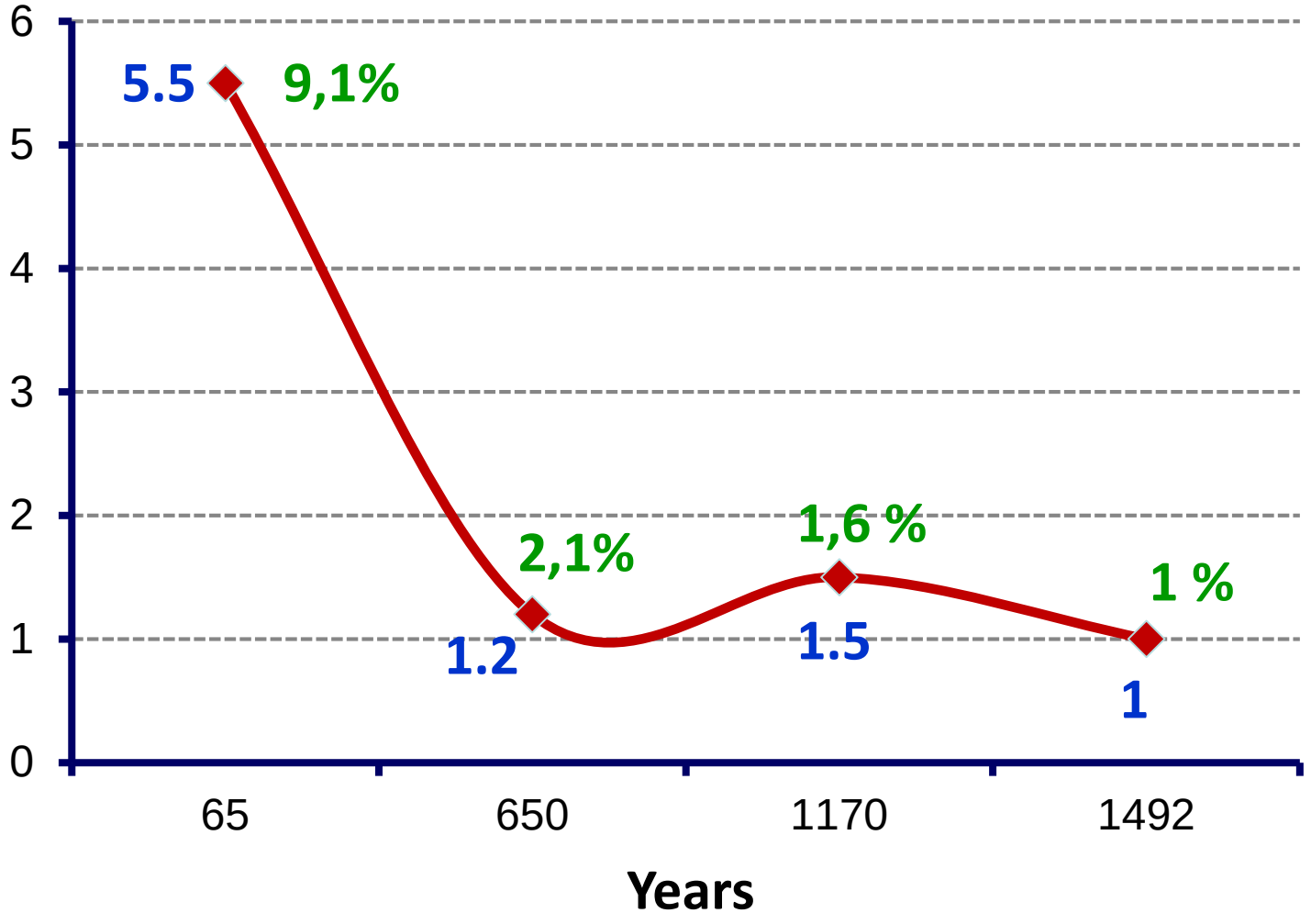
Jewish Population (in million), 70 CE - 1492



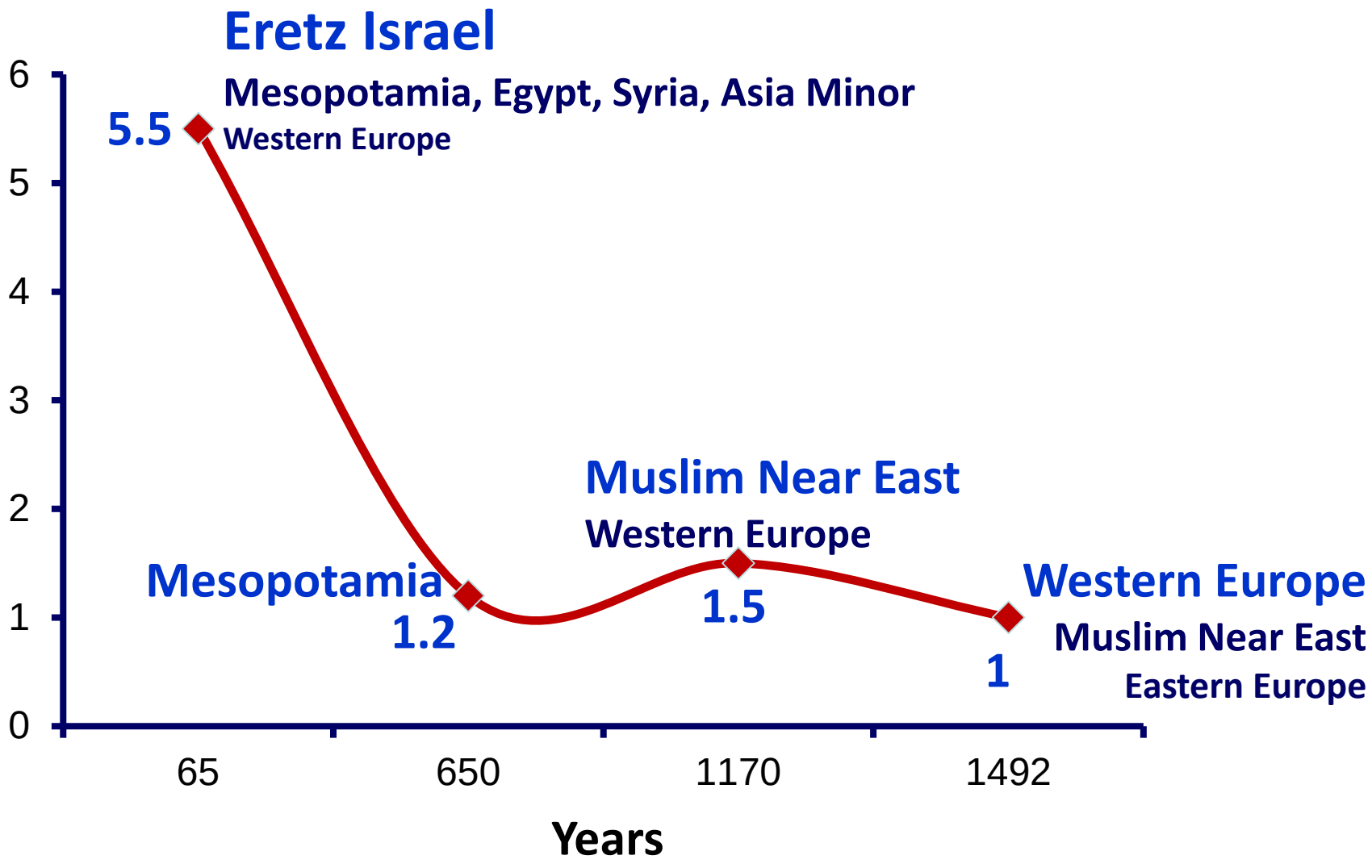
Total Population (in million), 70 CE - 1492



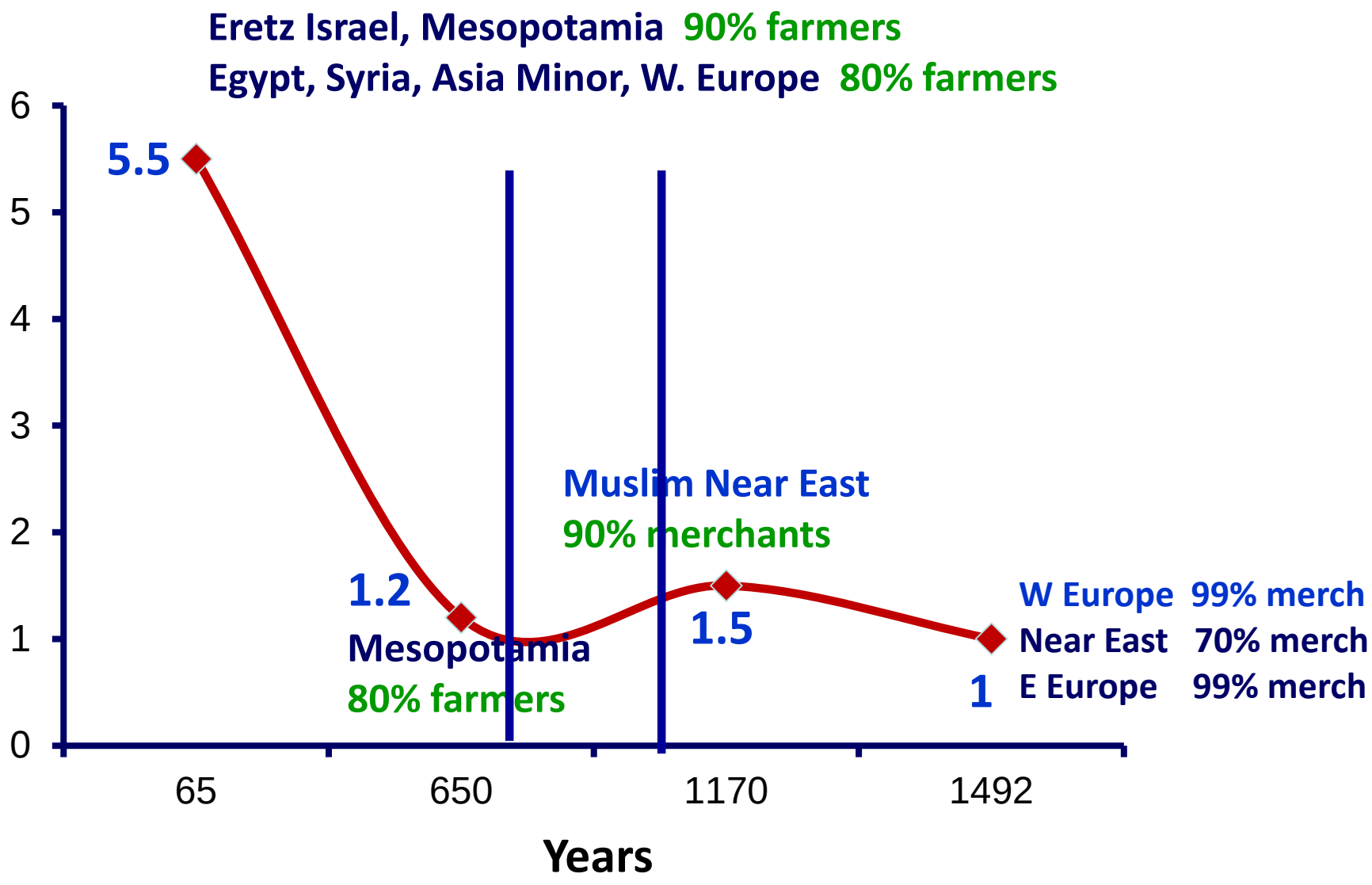
Jewish Population (%), 70 CE - 1492



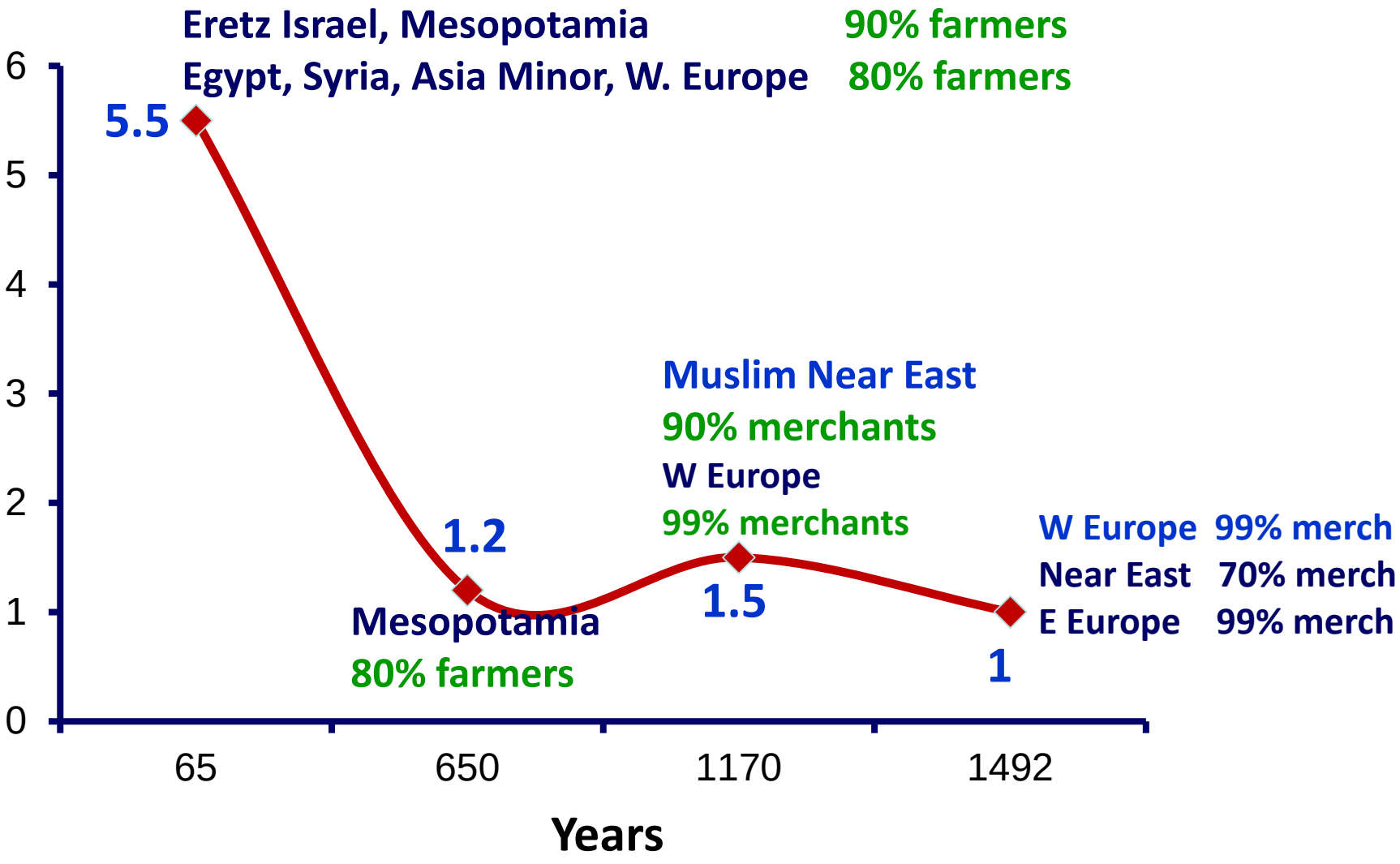
Centers of Judaism and Jewish Life, 70 CE - 1492



Jewish occupational transition, 750 - 900



Diaspora: the migrations of Jewish skills, 800-1500



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Why are there so few Jews in farming?

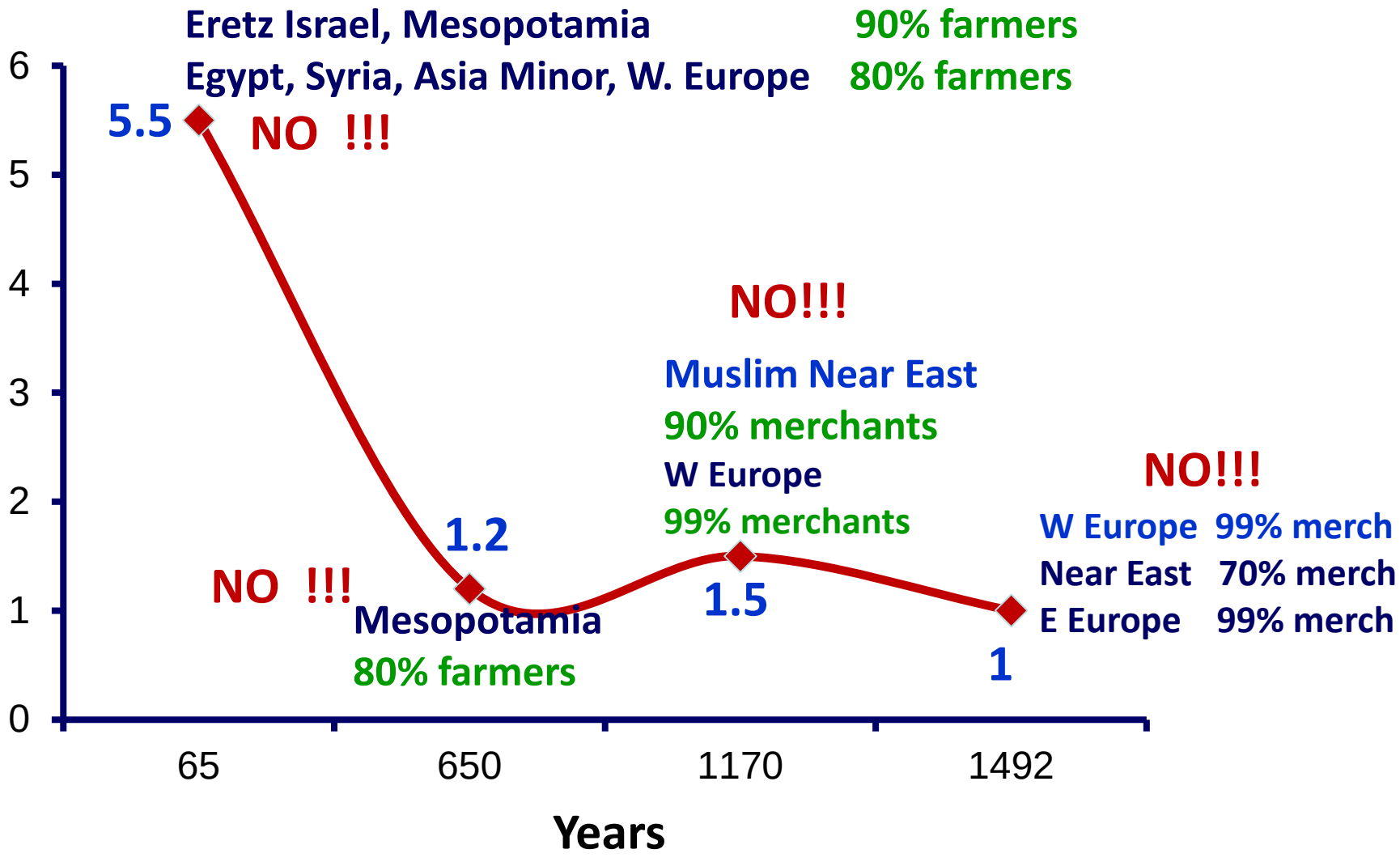
(… many traders, entrepreneurs, bankers, lawyers, doctors, profs…)

1. **Economic restrictions / prohibitions** (e.g., Cecil Roth)

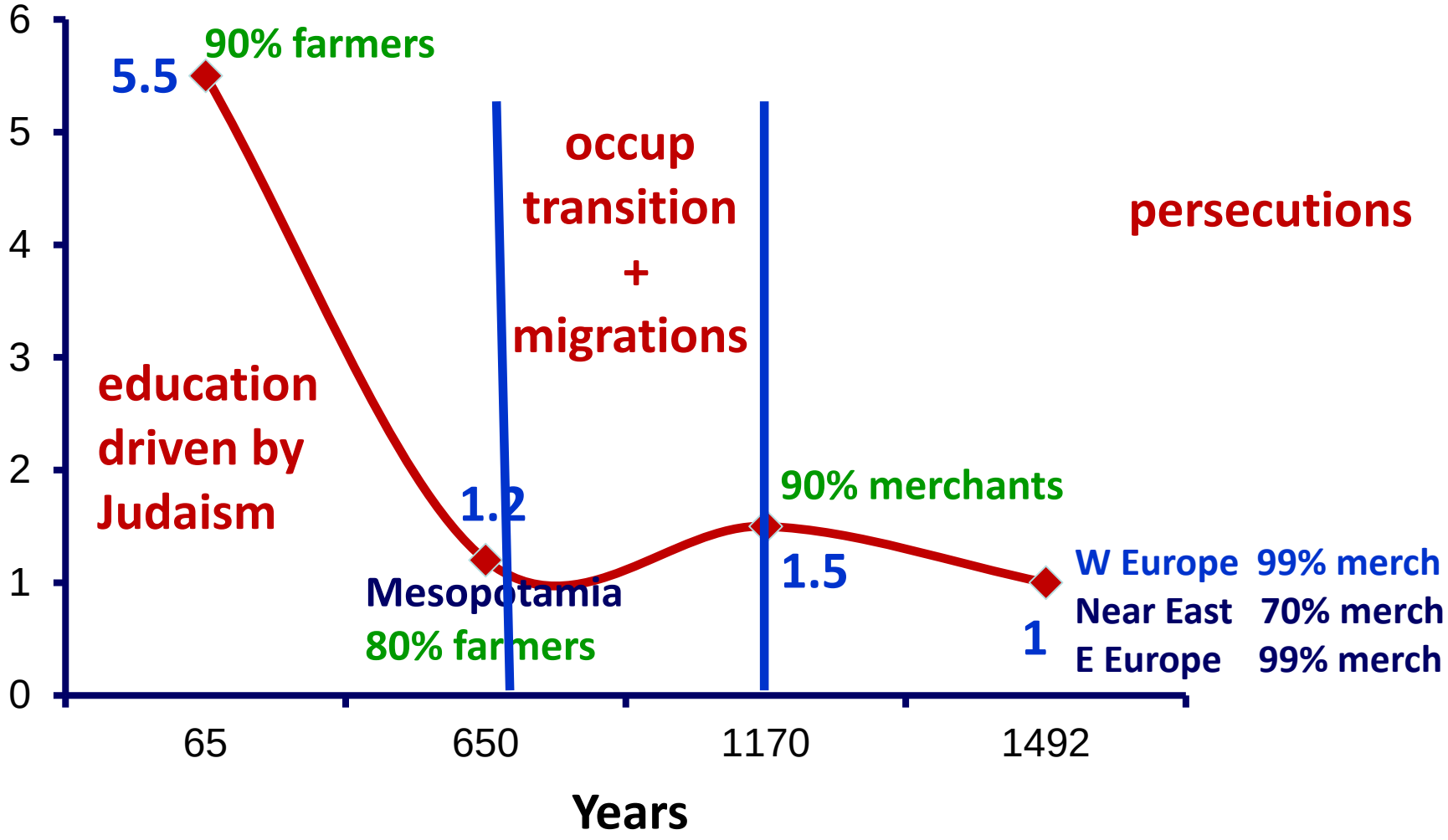
2. **Persecutions & portable human capital** (e.g., Brenner & Keefer)

3. **The economics of small minorities** (e.g., Weber; Kuznets; Slezkine)

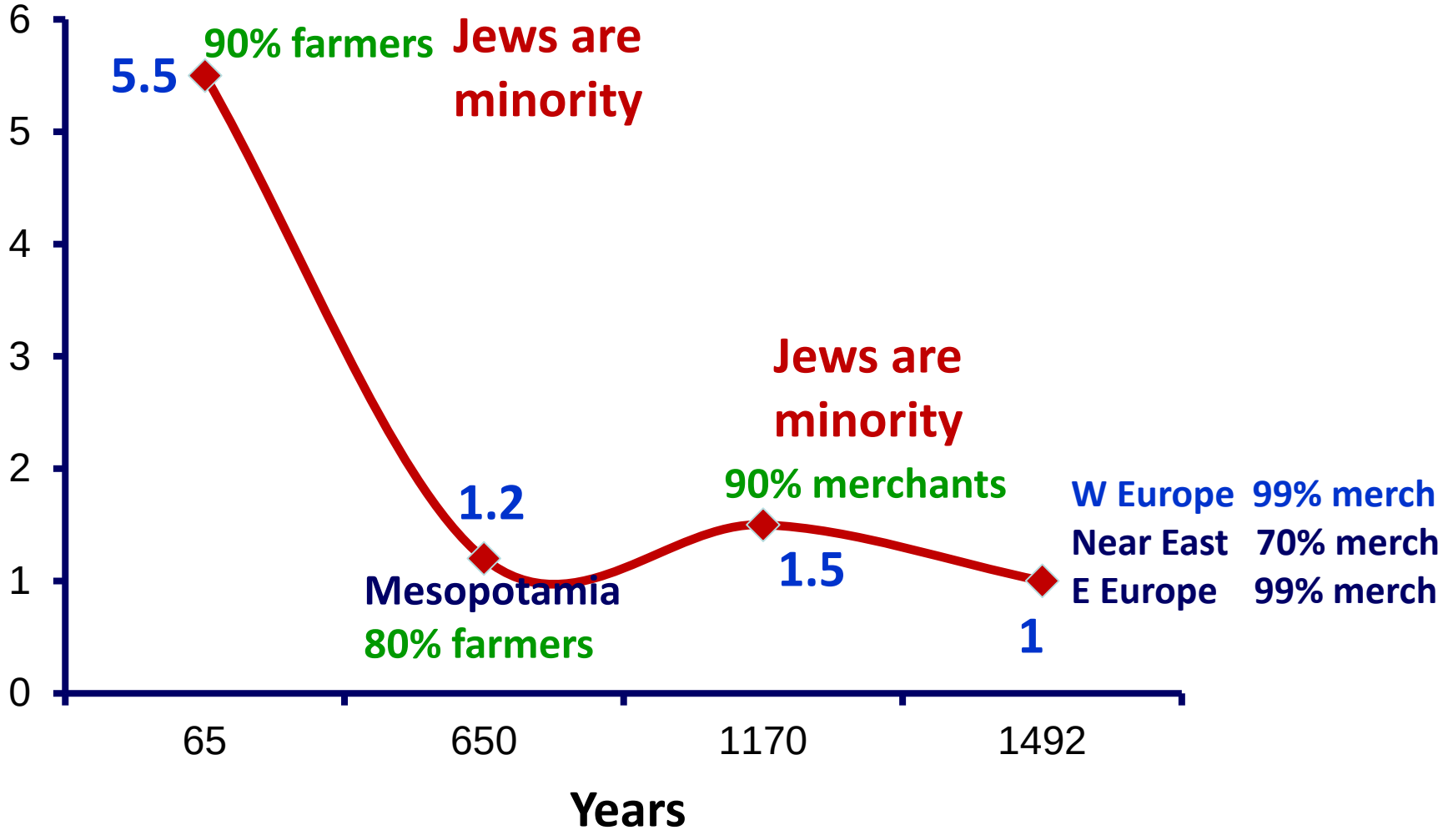
1. Jewish occupational transition: restrictions?



2. Jewish occupational transition: persecutions and portable human capital?



3. Jewish occupational transition: economics of small minorities?



The Chosen Few: why?

Is there a common factor
behind the three historical patterns / puzzles?

Our answer

- a shift in the **religious norms** within Judaism
- the development of communal **institutions** within Judaism

after 70 CE brought these long-term economic and demographic outcomes

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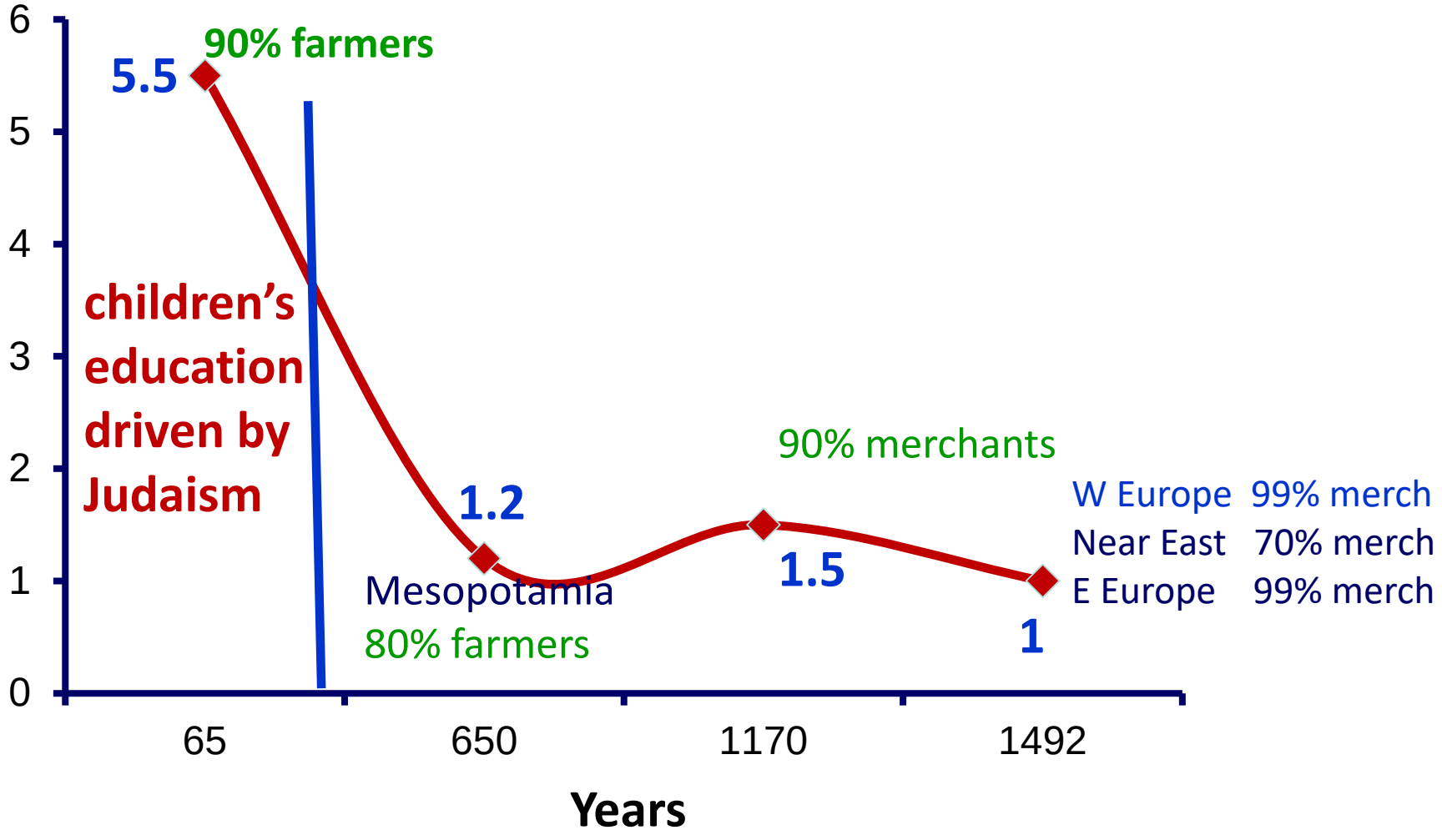
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Ch 10 1492 to today: open questions

First "historical accident", 70 CE

200 BCE – 70 **Temple and Torah**	70 **Torah**	70 – 200 **Torah**
Many religious groups Pharisees: stress the study of Written and Oral Torah (Law) About 64 Religious norm: fathers must send sons to school to study the Torah	Temple destroyed Pharisees became religious leaders Sacrifices replaced with study of the Torah in synagogue	Leadership of rabbis Mishna, Rabbinical Courts Communal Institutions From 200 *ammei ha-aretz* (illiterate people) considered outcast

First "historical accident": a "literate" religion



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Model

- Two-period overlapping generations model with no population growth
- 1st period: the child receives education, e_s
- 2nd period: the adult decides his religion (j, n) and child's education
- Before 200 CE: Jews and non-Jews have same level of education (= 0) and income
- After 200: change in Jewish religious preferences

Utility function



Jewish individual

$u^j(c, e_s; e, x) = log\ c + x(e+1)e_s - \varepsilon h$

Jewish individual who converts

$u^{jn}(c, e_s; e, x) = log\ c - \pi x$

Non-Jewish individual

$u^n(c, e_s; e, x=0) = log\ c$

$x\ (>0)$ exogenous taste parameter

"Literate" religion: interaction of x with e_s and e

- Cost of child's education $\gamma(e_s)^{\theta}$ $\gamma > 0, \theta > 1$
- At community level
  - operating costs of synagogue, teacher's salary, cost of books
- At individual level
  - child's cognitive ability / diligence
  - opportunity cost of time the child spends in school
- **Education does not affect productivity and earnings in farming**
- Budget constraint $c + \gamma(e_s)^{\theta} + \tau^{rF} \leq w^{F}$

Testable implications on children's education

$e_s^* = 0$ if $x(e+1) < (\gamma\theta) / (w^F - \gamma - \tau^{jF})$

$e_s^* \geq 1$ otherwise, and e_s^* solves the equation
$x(e+1) = (\gamma\theta(e_s)^{\theta-1}) / (w^F - \gamma(e_s)^{\theta} - \tau^{jF})$

- At the community level
  - γ large in small Jewish communities
  - negative shocks lower w^F
- At the individual level
  - families with high opportunity cost of sending children to school
  - fathers with low x or low e

Jewish farmers: conversion

A Jewish farmer converts if

$$log(w^F - \gamma(e_s^*)^\theta - \tau^{jF}) + x(e+1)e_s^* - \varepsilon h < log(w^F - \tau^{nF}) - \pi x$$

Suppose $\tau^{jF} = \tau^{nF}$

1. Farmers who do not educate their sons, convert if $\varepsilon \geq \pi x \geq 0$
2. Farmers who do not educate their sons, do not convert if $\varepsilon < \pi x$
3. Farmers who educate their sons, do not convert even if $\pi = 0$

Testable implications on conversions and Jewish population dynamics

At a given point in time:

- Because of heterogeneity across individuals (x, γ, θ, e), some Jewish farmers do not educate their children and convert

- More conversions occur when aggregate economic conditions are bad (low w^F, high τ^{rF}) and in small communities (high γ)

In the long-run, Judaism cannot survive in a subsistence farming society as the Jewish farming population will decrease through conversions

Reduction in Jewish population can be halted

1. with increased urbanization and the expansion of trade
2. with migrations

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Jews in the Talmud Era (200-650): The Chosen Few [children's education]

- In a subsistence farming economy, the investment in children's education is a religious sacrifice with no economic return

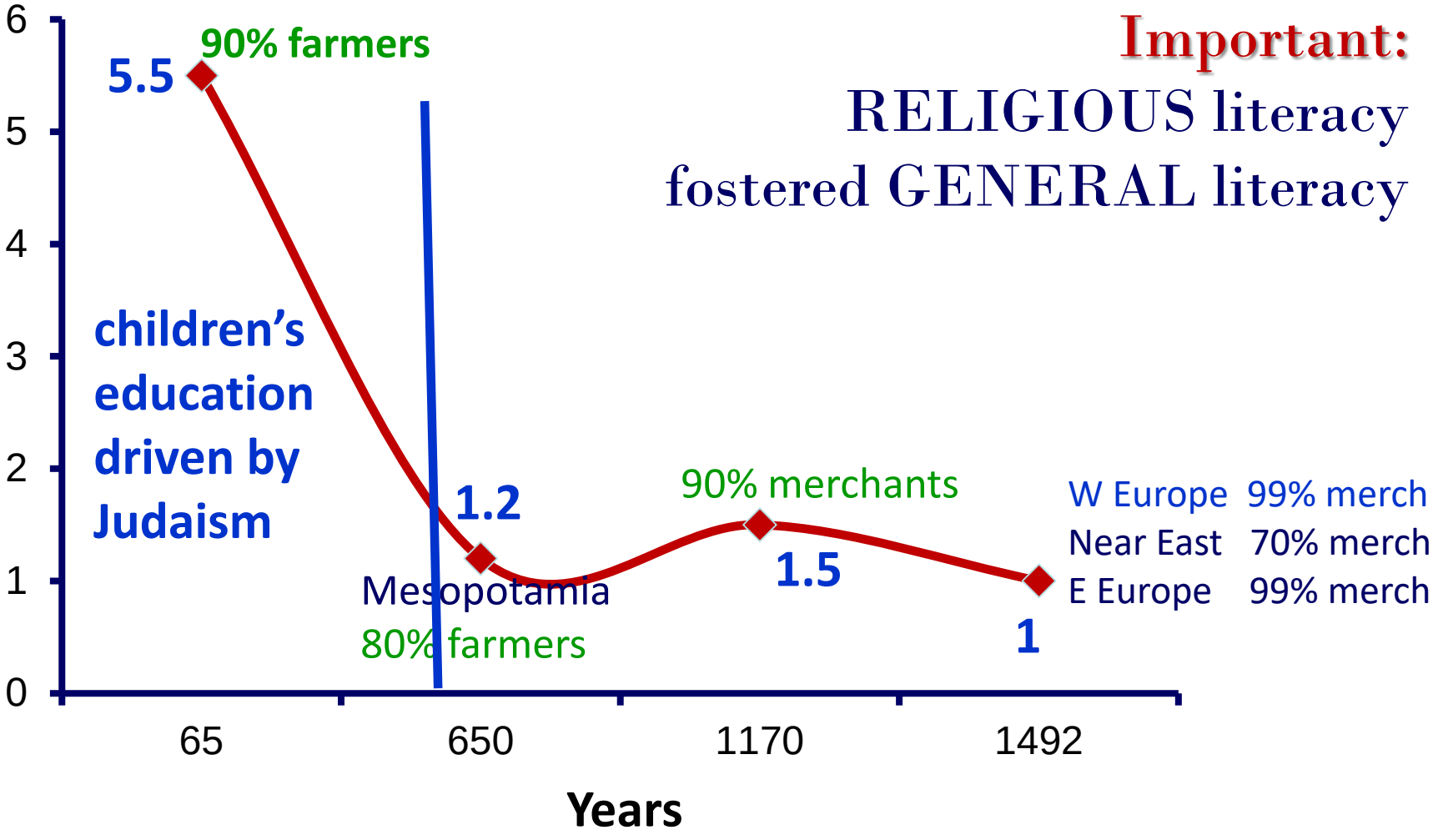
- A typical family's budget in Roman Palestine
  - food expenses = 40-50%
  - taxes = 30%
  - little was left to buy clothing, books, and paying for the teacher's salary

Cost of living (denarii), 1st-3rd centuries

Items in a household budget	Land of Israel	Egypt	Babylon
Monthly wage of agricultural worker	**24-48**	**4-32**	**72-96**
Monthly wage of urban skilled worker	**48-72**	**6-40**	
Monthy wage of boy on farm work	**---**	**2-10**	
Monthly bread expenses (family of four)	10-20	5-10	
Cattle (ox or cow)	100-200	15-100	
Suit/cloak	30	---	
Monthly rent of a house	4	---	
Book	**200**	**---**	**80-120**

Source: Sperber (1965; 1967)

Despite being costly, primary education became spread from 200 to 650 among Jews (farmers)

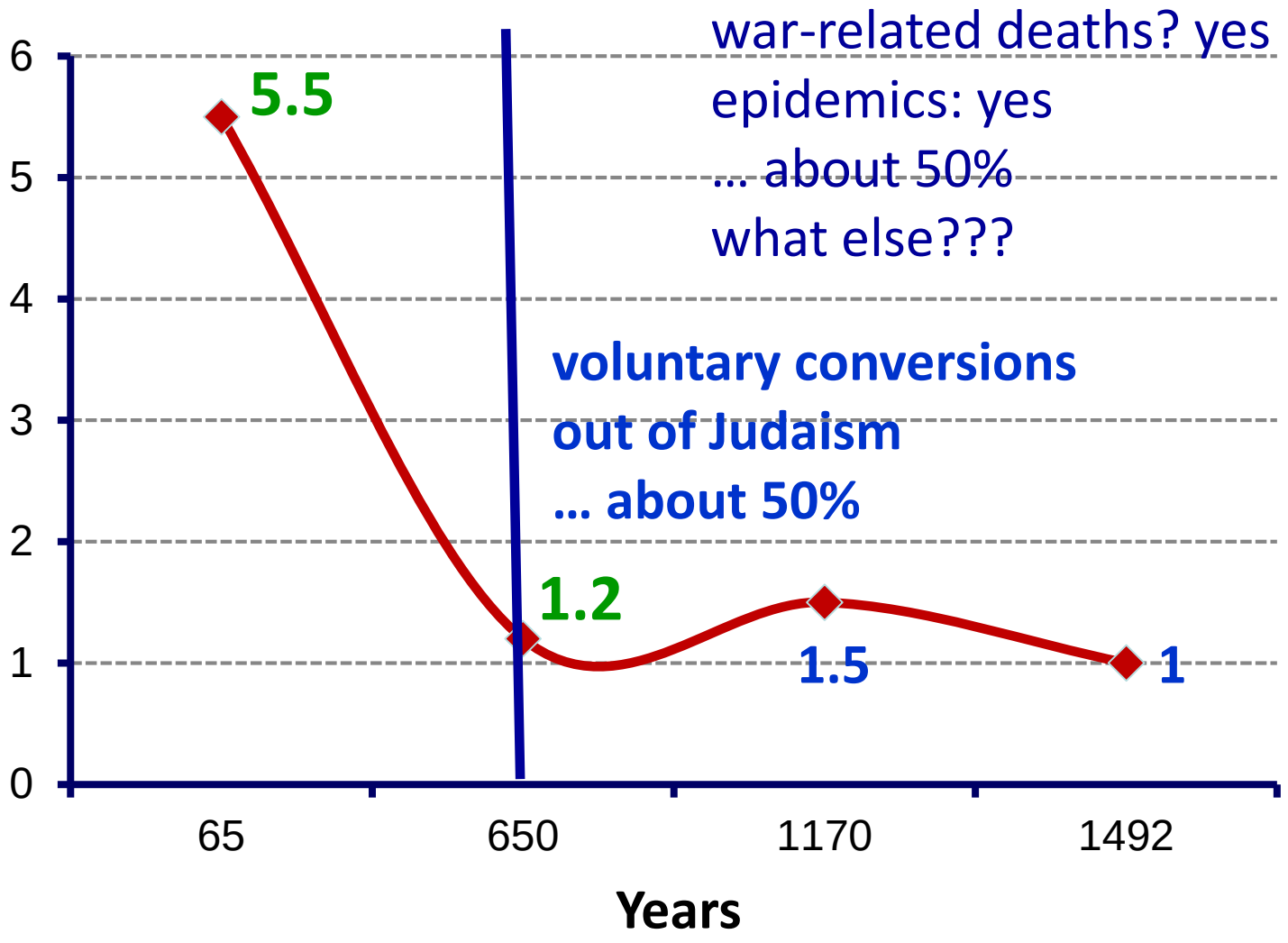


Important:
RELIGIOUS literacy
fostered GENERAL literacy

Jews in the Talmud Era (200-650):

The Chosen Few

The Chosen FEW: why?



Years	Emperor	Decree
311-337	Constantine	Death penalty for Jews who harm Jewish converts to Christianity
364-375	Valentinian	Jewish parents cannot disinherit children who converted to Christianity
379-395	Theodosius	Death penalty by fire for Jews who harm Jewish converts to Christianity
395-423	Honorius	Jewish converts to Christianity can revert to Judaism
395-408	Arcadius	Jews cannot become Christians for economic motives
527-565	Justinian	Jewish parents cannot disinherit children who converted to Christianity

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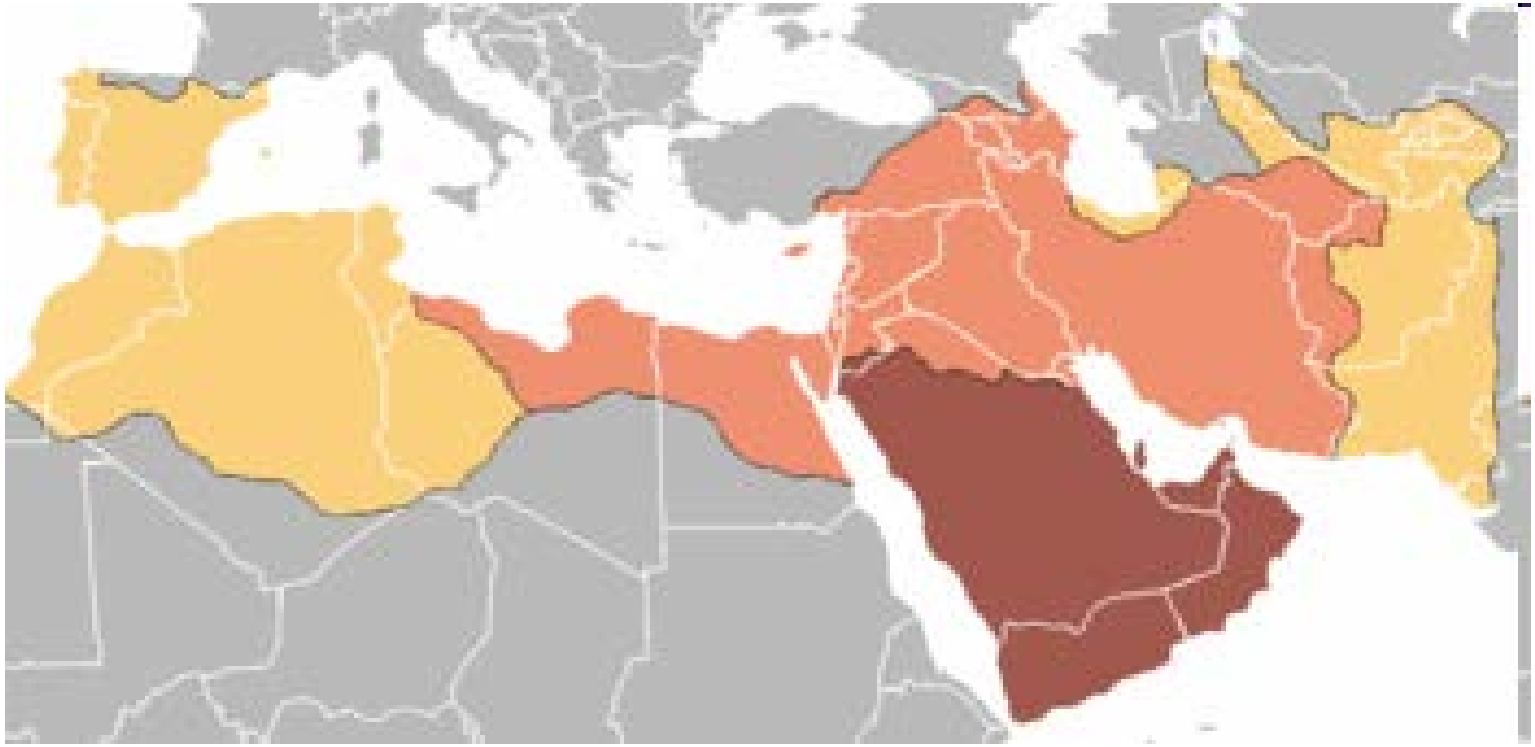
Ch 10 1492 to today: open questions

If most Jews were now literate, why were they still mostly in farming around 650?

As economies in 4th-7th centuries were still mostly agrarian, (literate) Jewish farmers could not find urban skilled occupations that required / benefited from, literacy

Second "historical accident", c. 622

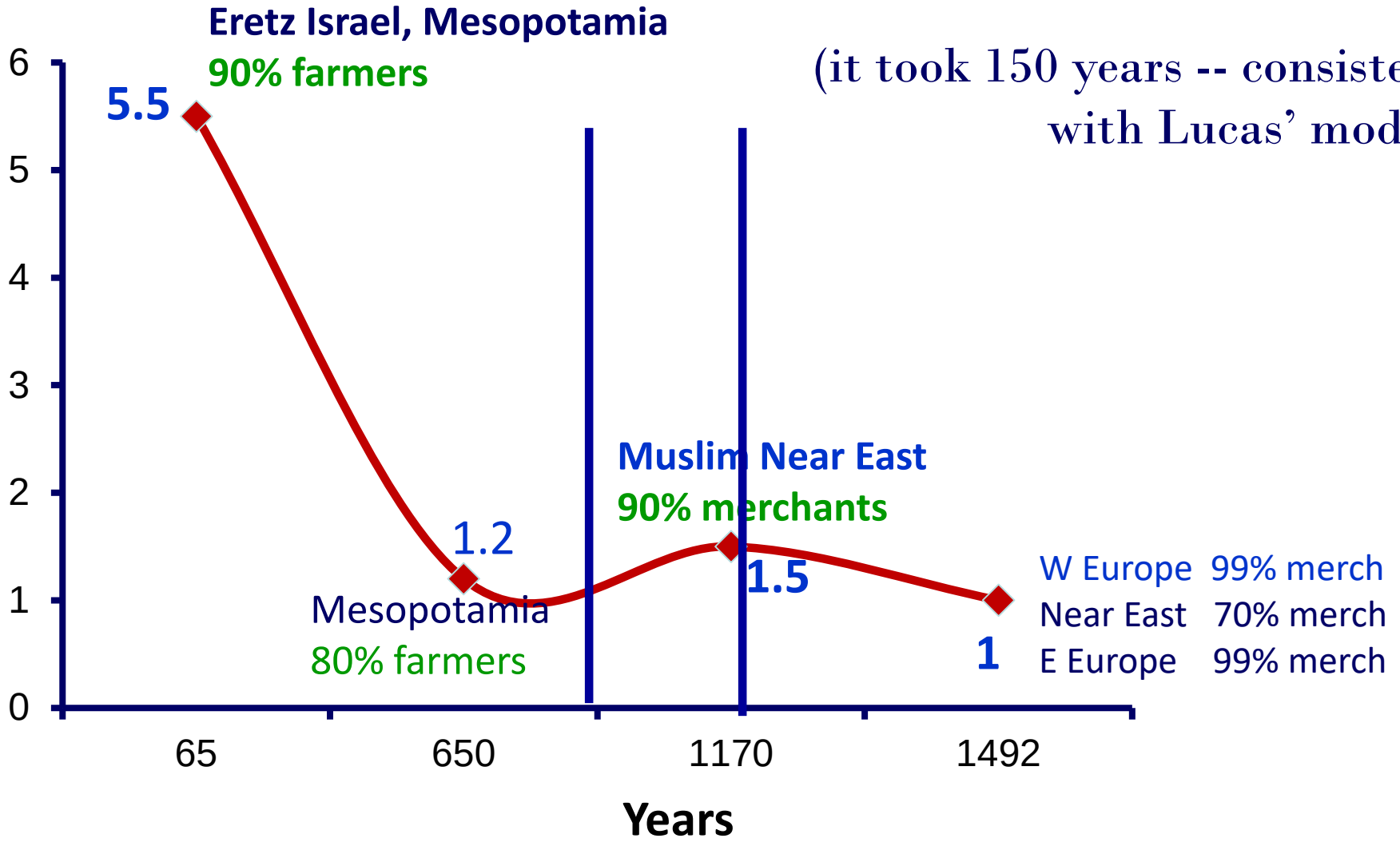
Mohammed established Islam and set the foundations for one of the largest, most urban, commercially developed, and scientifically advanced, empires in history



Urbanization expanded in newly established Abbasid Empire

8th – 9th centuries	Total Population (thousands)
Baghdad	600—1,000
Samarra	500
Basra	200-600
Cairo	300
circa 1170	
Palermo	150
Paris	110
Seville	80
Venice	70
Granada	60
Cordoba	60

Jewish occupational transition, 750 - 900



(it took 150 years -- consistent with Lucas' model)

The model for merchants

Merchant's budget constraint

$$c + \gamma(e_s)^{\theta} + \tau^{rM} \leq w^{F}(1 + A e_s^{\alpha} e^{1-\alpha})$$

Education

- Jewish merchants invest more than non-Jewish merchants in children's education. Why?

Conversion

(i) If $\tau^{jM} = \tau^{nM}$, no Jewish merchant will convert

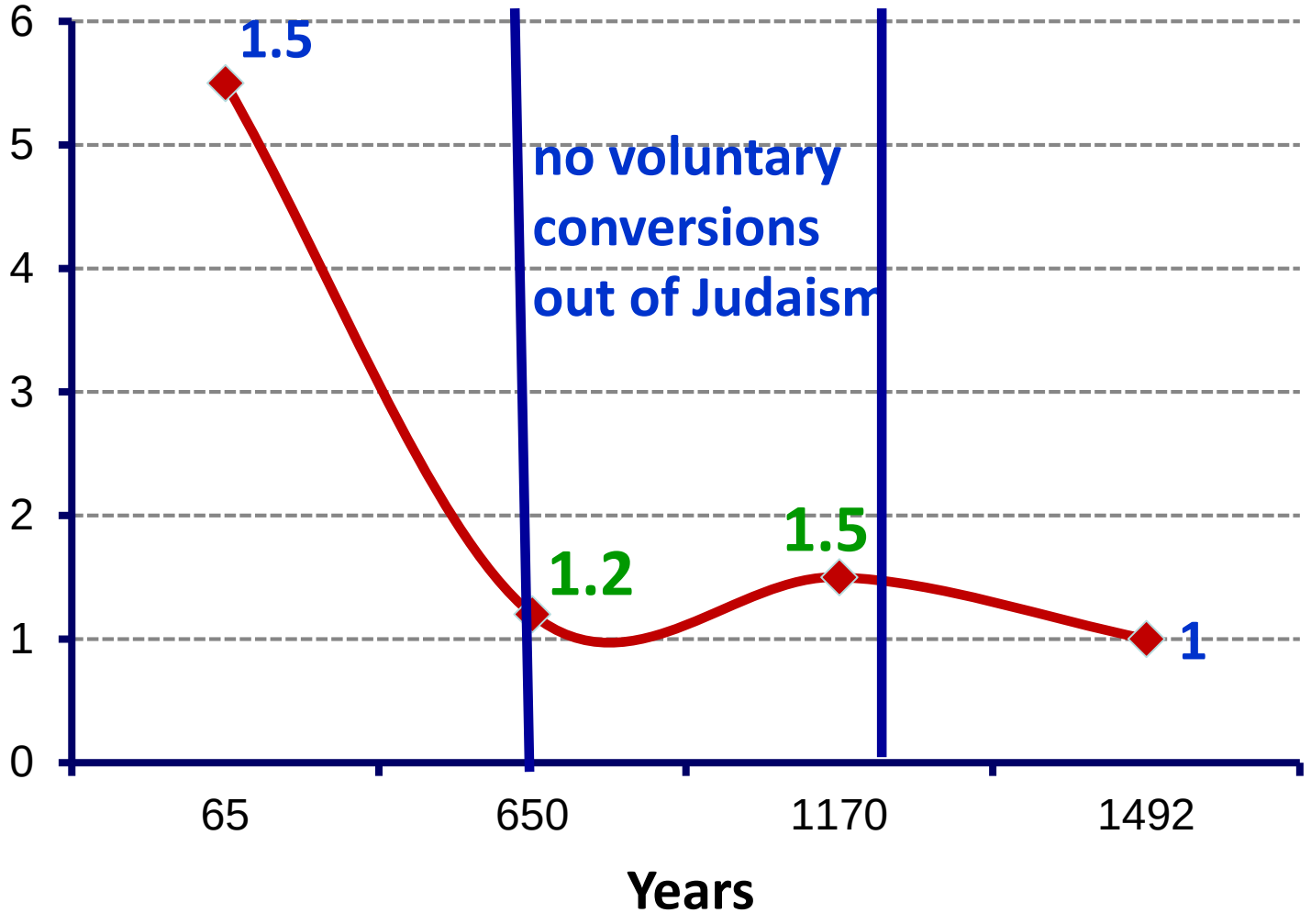
(ii) Over time, the proportion of Jews in urban skilled jobs (e.g., merchants) will increase

Education: tons of evidence from Genizah and *Responsa*

No or few conversions of Jews from 700 to 1200

Jewish Population Dynamics		
	c. 650	**c. 1170**
Land of Israel	0.1	0.002
Mesopotamia and Persia	**0.7-0.9**	**0.8-1.0**
Egypt and North Africa	0.004	0.07
Syria	0.005	0.015
Balkans, eastern Europe	0.047	0.047
Western Europe	0.005	0.103
Total Jewish Population	**1-1.2**	**1.2-1.5**
Total Population	**51.1**	**70**
J as % of total pop	**2.1%**	**1.6%**

Jewish Population Dynamics, 650 - 1170



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Voluntary Diaspora

Migrations of Jewish *skills*, ca. 800-1250

- **Main insight from the model**

  Judaism can survive in the long run only if Jews can find occupations with high returns to their investment in education

- **Historical evidence**

  The voluntary migrations of Jewish people between 800 and 1250 support this argument

Migrations within the Muslim Empire (800-1100)
voluntary and free

- Jewish craftsmen, traders, physicians, scholars from Mesopotamia and Persia settled in Syria, Egypt, Maghreb, Spain, and Sicily
- The (first) "golden age" of Jewish history

Migrations to western Europe (850-1250)
voluntary and regulated

- Jews migrated to England, Flanders, France, Germany, Italy upon invitation by local rulers --- wealthy communities in hundreds of towns
- Because of high human capital and skills, Jews viewed as essential for economic growth
- No restrictions on Jewish economic activities

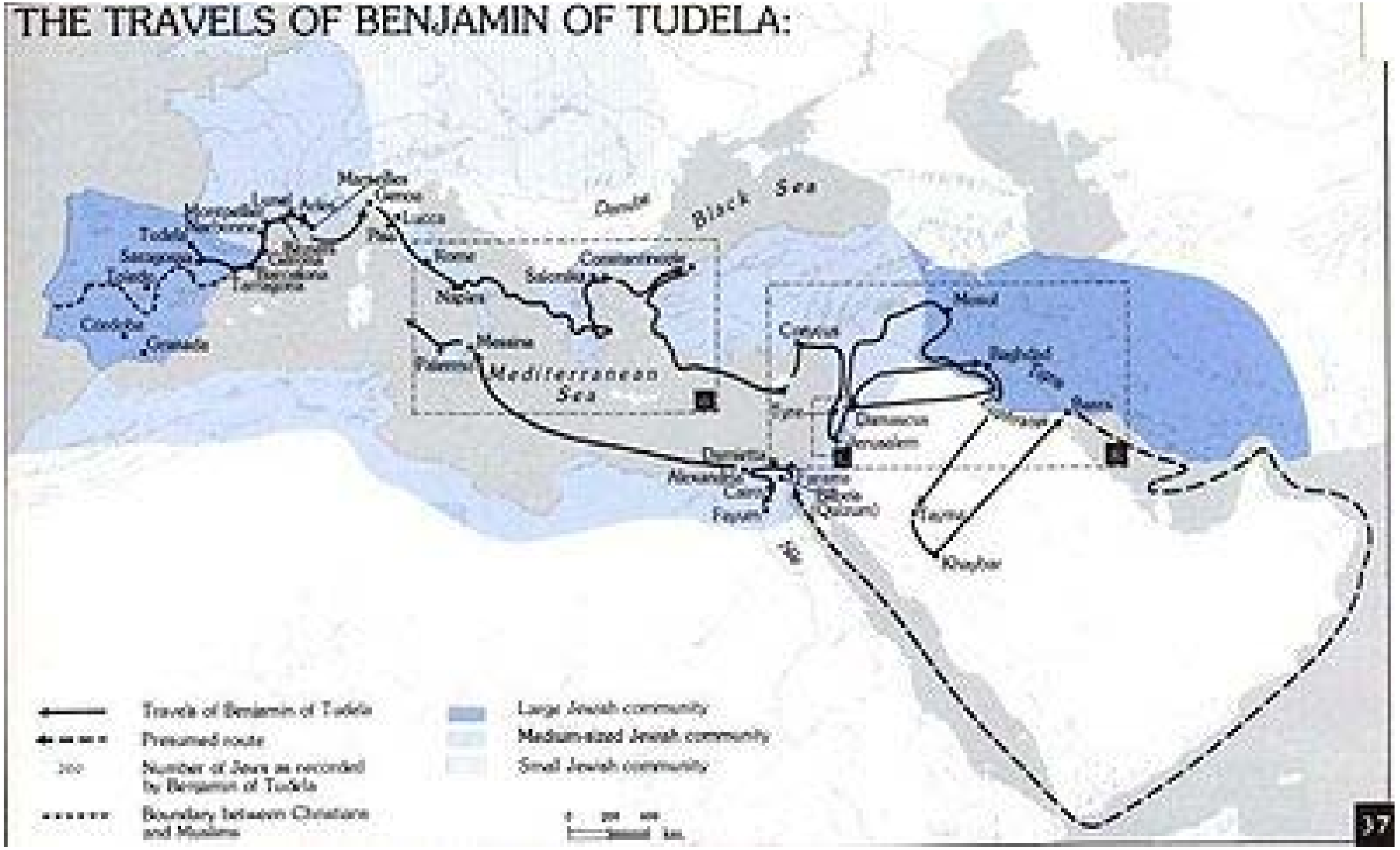
Sample of Medieval Charters

Country	City	Year of charter	Own Land	Trade	Money Lending
Spain	Barcelona	1053-1071	yes	yes	yes
	Tudela	1116	silent	yes	yes
	Toledo	1222	yes	yes	yes
	Valencia	1250	yes	yes	yes
France	---	820	yes	yes	silent
	---	1190	silent	silent	yes
England	---	1120, 1170	yes	yes	yes
	---	1275	yes	yes	no
Germany	Speyer	1084, 1090	yes	yes	yes
	Worms	1074	silent	yes	silent
	Worms	1090, 1157	yes	yes	yes
	Ratisbon	1182, 1216, 1230	yes	yes	silent

The zenith of the Jewish Diaspora

From the travel itinerary of Benjamin de Tudela (**c. 1170**)

- **In Muslim Mesopotamia and Persia: 70 percent of world Jewry**
- **Muslim Iberian Peninsula**: wealthy Jewish communities in hundreds of cities and towns (*Sephardim*)
- **France, England, Germany**: prominent Jewish communities in hundreds of locations (*Ashkenazim*)
- **Jewish communities all over** Italy, Bohemia, eastern Europe, Turkey, the Middle East, Egypt, the Maghreb, all the way to central Asia, China, and India


THE TRAVELS OF BENJAMIN OF TUDELA:
Black Sea
Mediterranean Sea
Travels of Benjamin of Tudela
Presumed route
Number of Jews as recorded by Benjamin of Tudela
Boundary between Christians and Muslims
Large Jewish community
Medium-sized Jewish community
Small Jewish community
37

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Timing of Specialization in Moneylending

- **Circa 1000:** main occupations of Jews in Spain, France, Germany, and southern Italy were local commerce, long-distance trade, crafts, and medicine
- **Circa 1100:** moneylending was *the* occupation of Anglo Jewry; a very important occupation of French Jewry; one of many professions of the Jews in Germany, the Iberian Peninsula, and southern Italy
- **By 1300:** almost all Jews in France, Germany, and northern and central Italy were engaged in moneylending (unlike Spain and southern Italy)

Why? Economic Restrictions ?

- **Most popular view, I**

  Jews in medieval Europe became segregated in moneylending because they could not own land (no farming) & guilds pushed them out of urban skilled occupations

- **Most popular view, II**

  Usury bans on Muslims and Christians led to Jewish specialization in moneylending

Jewish Land Ownership ?

Country	City	Year of charter	Own Land	Trade	Money Lending
Spain	Barcelona	1053-1071	yes	yes	yes
	Tudela	1116	---	yes	yes
	Toledo	1222	yes	yes	yes
	Valencia	1250	yes	yes	yes
France	---	820	yes	yes	---
	---	1190	---	---	yes
England	---	1120, 1170	yes	yes	yes
	---	1275	yes	yes	no
Germany	Speyer	1084, 1090	yes	yes	yes
	Worms	1090, 1157	yes	yes	yes
	Ratisbon	1182, 1216	yes	yes	---

Usury Bans & Guilds ?

Time	Location	
650-1250	Muslim Empire	Koran prohibits Moslems from charging interest
750 – 900	Mesopotamia and Persia	Jews left farming and entered nearly 450 occupations (crafts, trade, moneylending)
850-1250	Europe	Jews migrated from Byzantine Empire and Muslim Middle East to Europe as urban dwellers specialized in crafts, trade, and moneylending
From 1100	Europe	Jews became prominent in moneylending Jewish scholars (e.g., Rashi) issued rulings to regulate moneylending during 11th and 12th cent
1200-1350	Europe	Church strictly enforces usury ban on Christians
1200-1350	Europe	Crafts and merchant guilds began growing
1350-1550	Europe	Guilds dominated manufacturing and commerce
1350-1500	Europe	Restrictions on Jewish lending on mortgage

Our Explanation

- **Simple theory of occupational choice** (Roy's model): individuals choose an occupation based on (i) their advantage in the skills required for that occupation and (ii) the expected earnings in that occupation versus alternative occupations

- Trade and moneylending require more capital, more networking, more literacy and numeracy, and stronger contract-enforcement institutions than farming

- Prominence of Jews in financial intermediation is explained by their comparative advantages in the four assets that were (and still are) the pillars of financial intermediation

Jewish Specialization in Financial Intermediation: Segregation or Choice?

- It was not the outcome of restrictions and persecutions

- Jewish comparative advantage in the four assets required for successful financial intermediation led to





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Ch 3 The people of the book (c. 200 BCE — 200 CE)

Ch 4 The economics of Hebrew literacy in a world of farmers

Ch 5 Jews in the Talmud era (200-650 CE): the chosen few

Ch 6 From farmers to merchants (c. 750-900)

Ch 7 The educated wandering Jew (c. 800-1258)

Ch 8 From merchants to moneylenders: selection or segregation?

Ch 9 The Mongol shock: Can Judaism survive when trade and urban economies collapse?

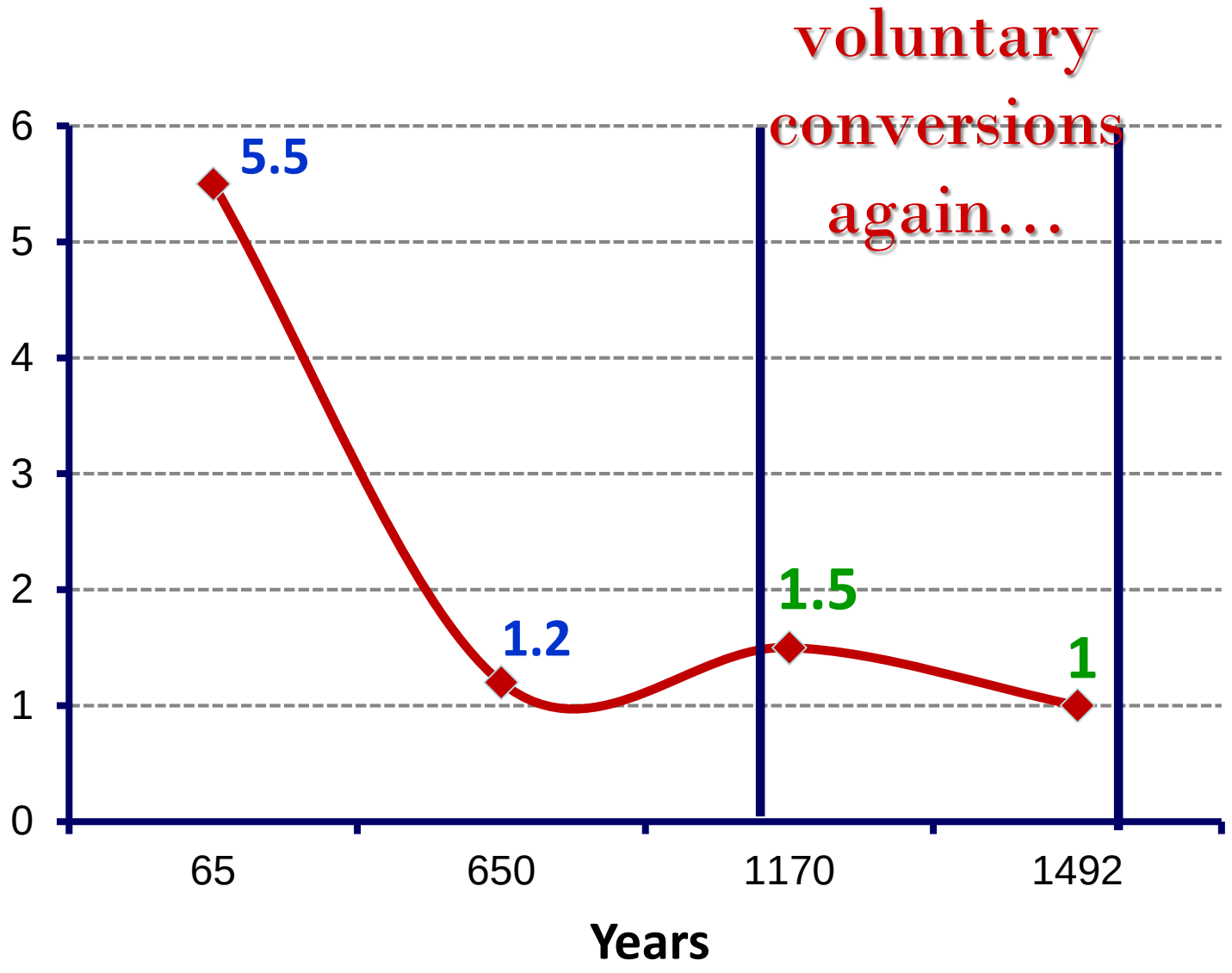
Ch 10 1492 to today: open questions

Third "Historical Accident", 1258
The Mongol Shock

(Could Jews be farmers in the long-run?)

- The Mongols invaded Persia (earliest 1220) and Mesopotamia in 1256-1260 and destroyed the urban economy
- Because of massacres, starvation, epidemics, total population was reduced by half
- Jewish population shrank from about 800 thousand to nearly 200-300 thousand

Jewish Population (in million), 70 CE - 1492



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Ch 10 1492 to today: open questions

1492 to Today: Open Questions

- **Circa 1492 world Jewry: about 1 million people**
  - 500,000 Sephardim (urban skilled occupations)
    Spain, North Africa, Greece, Turkey, Iraq, Iran
  - 500,000 Ashkenazim (urban skilled occupations)
  - Germany, Netherlands, Italy, eastern Europe, Russia

- **Circa 1938 world Jewry: about 16.5 million**
  - 2.2 million Sephardic Jews
  - 14.3 million Ashkenazi Jews (growth in eastern Europe)

- **Why this divergent demographic trend?**

1492 to Today: Open Questions

- Jews make 0.2 percent of the world population, and …
  - 54 percent of the world chess champions
  - 27 percent of the Nobel physics laureates
  - 31 percent of the medicine laureates

  Jews are 2 percent of US population, and …
  - 21 percent of the Ivy League students bodies
  - 26 percent of the Kennedy Center honorees
  - 37 percent of Academy Award winning directors
  - 38 percent of those on a recent *Business Week* list of leading philanthropists
  - 51 percent of the Pulitzer Prize winners for nonfiction
- Why this persistence in economic and intellectual success?

1492 to Today: Open Questions

Nowadays, world Jewry is about 16 million people

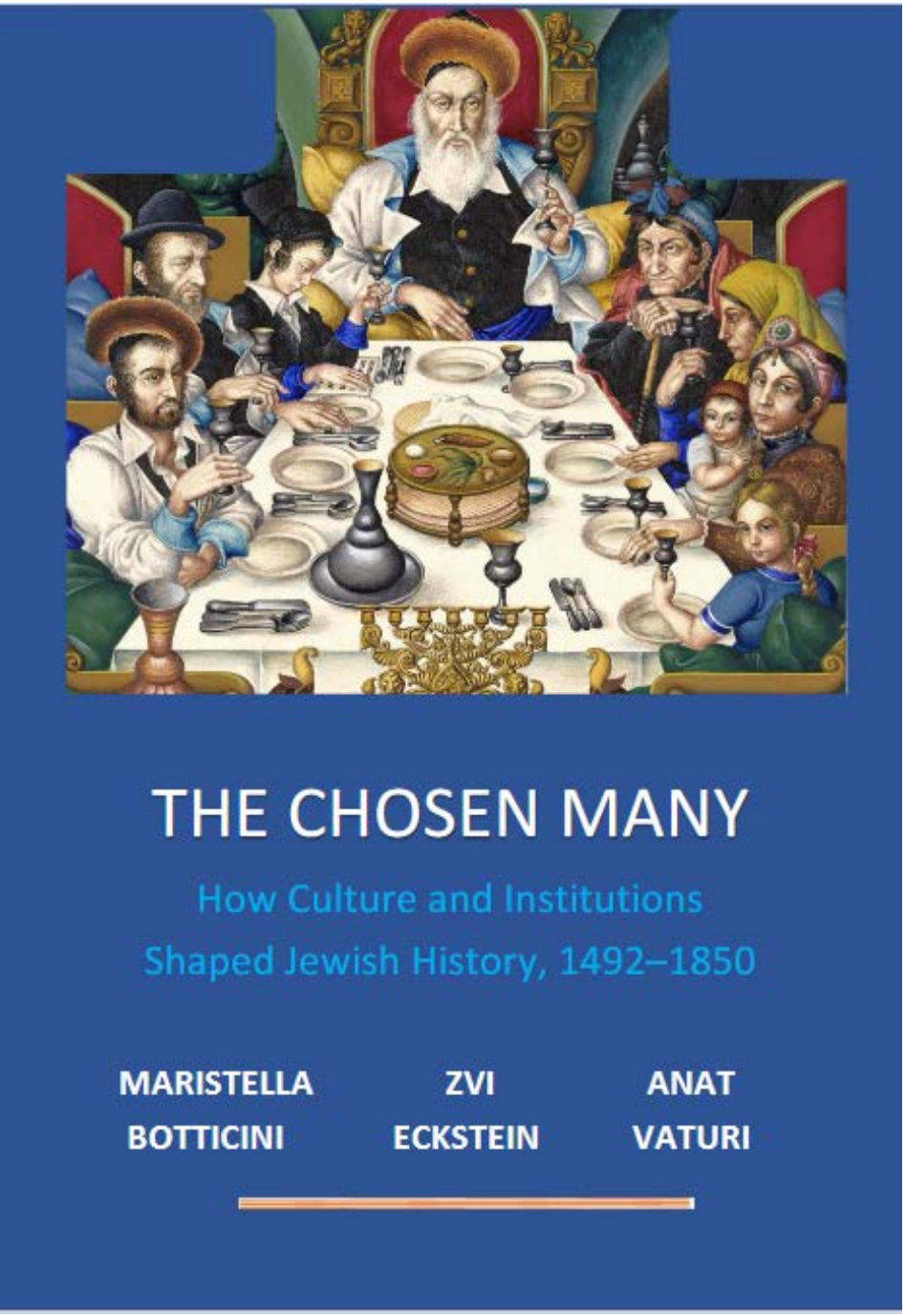
47% in the United States (A)

6% in the rest of the world (A)

47% in Israel (B)

– Jews in (A) display occupational selection (high-skill jobs) and have higher earnings than the rest of the population
– Jews in (B) have occupational structure similar to that of any small European country or that of the general population of the United States

- Why this different occupational and earning structure?



The Chosen Many

How Culture and Institutions Shaped Jewish History, 1492-1850

from

1 million c. 1492

to

5 million c. 1850

specialized in skilled jobs

Map 1
Poland-Lithuania circa 1650



PART I
Culture Matters: The Nexus Between Infant Care and Children Education

1. From Few to Many: Jewish Demographic Growth in Eastern Europe
2. Engines of Jewish Population Growth in Eastern Europe
3. Infant and Child Care: Medical Science and Religious Norms
4. The 'Chosen Healthy': Economic Analysis Meets Jewish History

PART II
Institutions Matter: The Economics of Nobility and the Community

5. The Republic of Nobles and the Economics of Nobility, 1450–1772
6. If It Was So Bad, Why Was It Actually Good for Jews, 1450–1772?
7. On the Road to the Economics of Modernity, 1772–1850

EPILOGUE
From Shtetl to New York and Tel Aviv: Open Questions in Jewish History

Ch. 1: From Few to Many: Jewish Demographic Growth in Eastern Europe

- Jews in Poland-Lithuania (PL)

  1500 12.5 thousand = 0.13% of the local population

  1850 4.6 million = 15.4% of the local population

  61% of world Jewry

  1500–1850 average annual growth rate **1.56%**

  the **highest in Europe**!

- Jews more literate than local populations
- Jews specialized in skilled occupations (commerce, moneylending, banking, crafts, leaseholding, medicine, law, real estate, etc.)

Ch. 2: Engines of Jewish Population Growth in Eastern Europe

- The drivers of this exceptional demographic growth?
  - Conversions?
  - Migrations? from West? from East?
  - Higher fertility?
  - Lower mortality?
- Why more literate and educated?
- Why specialized in skilled occupations?

Conversions into Judaism?

NO

The Origins of PL Jewry: migrations from the West? YES

- 9th-10th centuries: Jews of Germany-Austria (GA) came from southern Italy and France and established Ashkenazi communities through charters of privileges
- 12th century: Jews established first communities in Poland
- In 1264 they were granted the first princely charter of privileges based on German-Austrian charters.
- **14th centuries (post-Black Death and expulsions): sizeable immigration of Jews from GA to PL**
- Evidence from names, privileges, trade, language, communal organization and religious practices link the Jews in PL to the Jews in GA.

The Origins of PL Jewry: migrations from the East? NO

- **What about the Khazarian hypothesis?**
  - No evidence of their presence from privileges and taxes after 1500.
  - Other ethnic Jews (Sephardi and Karaites) continued to keep their distinctive religious practices within the Jewish community.

- **Conclusion: no unambiguous evidence for this potential source**

Higher fertility? NO

Lower mortality? YES, *but* …

70% of higher Jewish population growth rate in PL was due to **higher infant and child survival rates**

… but WHY ???

Table 2.1: Births, deaths and natural increase per 1000 in Poland-Lithuania (Galicia and Posen), 1777-1873

Years	Place	Births Jews	Births Total Pop	Deaths Jews	Deaths Total Pop	Natural increase Jews	Natural increase Total Pop
1777-1800	Galicia	36.5	35.4	20.2	25.4	16.5	10.0
1824-1873	Posen	35.3	41.1	22.2	30.0	13.1	11.1
Average		**36**	**38**	**21**	**28**	**15**	**10**

Why Jewish death rate was lower?
Infant and child mortality: the main driver

Table 2.2: Infant and child mortality rates per 1,000 live births (Province of Posen in 1819-1863)

Age at death	Jews	Total pop	Percentage difference
Just before or at birth (A)	24	29	-17
Under 1 year old (B)	146	200	-27
All under 1 year old (A+B)	170	229	-26
1-3 years old	91	107	-15
3-5 years old	33	47	-30
All 1-5 years old	124	154	-20
All 0-5 years old	294	383	-23

Infant Mortality Rates per 1,000 live-born in PL and GA

	Jews	Total pop	Percentage difference
Until 1870	150	223	-33
1870-1910	155	279	-44
1910-1930	58	145	-60

- Lower child mortality up to age 5 accounts for 70% of lower death rate among Jews

70% of higher Jewish population growth rate in PL was due to **higher infant and child survival rates**

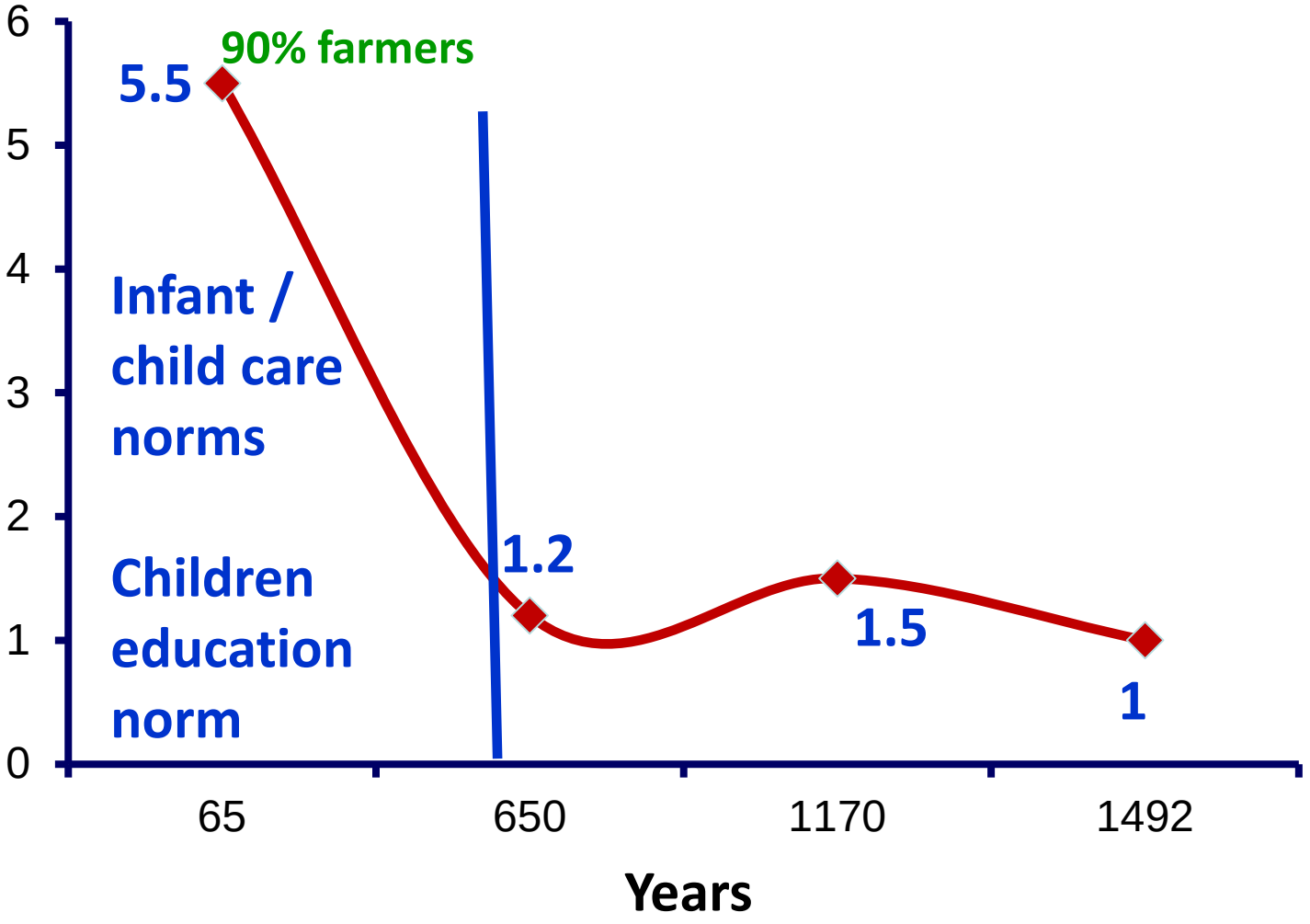
… but WHY ???

Ch. 3: Infant and Child Care: Medical Science and Religious Norms

- Today medical science prescribes a set of recommendations / best practices to enhance infant and child health and increase survival rates.
- **Jewish religious norms and practices regarding infant and child care** dating back to Talmudic (2nd – 5th centuries) times basically match most of these medical recommendations.
- Critical role of **communal institutions** (e.g., rabbis) in spreading and enforcing these norms and practices.

In doing so, they contribute to higher infant and child survival rates (--- and through this channel to Jewish exceptional population growth rate in PL).

Judaism: norms and practices



Ch. 4: The 'Chosen Healthy': Economic Analysis Meets Jewish History

- … ok. But: ***why*** Judaism had these norms and practices regarding infant and child care?

Our answer

- **Parental investment in children's education** prompted by Jewish religious norm (Talmudic times) **went hand in hand with parental investment in infant and child care norms and practices** prompted by Judaism (Talmudic times)

- **New in our book:** link between children's education and infant/child care

Ch. 4: A Model of Parental Investment in Infant and Child Care

- **Human capital theory** (Becker 1962)
  - Investment in children education entails costs and benefits
- **Parental investment in early childhood care** (Heckman 2006)
  - Investment in early childhood care is the key
- **Economics of social capital**
  - Rabbis as information sharing-mechanism
  - Communal institutions helped support both infant and child care norms and practices and children education within the community

Part II Institutions Matter: The Economics of Nobility and the Community

5. The Republic of Nobles and the Economics of Nobility, 1450–1772

- Early modern PL: very peculiar political and economic institutions (vs western Europe)
- Republic of nobles (vs monarchy)
- Economics of nobility --- monopoly of all production inputs (land, labor, capital)
  - Manorial system
  - Leaseholding
  - Private towns

5. The Republic of Nobles and the Economics of Nobility, 1450–1772

Economics of nobility

- Nobles needed assets and skills
  - money / loans
  - managerial skills to manage their properties (latifundia, alcohol production and sales, markets in private towns, etc.)
  - literate people who could read and write contracts and do math (accounts)
  - networking abilities to move goods and capital across locations

6. If it was so bad, why was it actually good for PL Jews, 1450–1772 ?

Who had those assets and skills needed

by the PL nobility? Jews

- money / loans
- managerial skills to manage their properties (latifundia, alcohol production and sales, markets in private towns, etc.)
- literate people who could read and write contracts and do math (accounts)
- networking abilities to move goods and capital across locations

6. If it was so bad, why was it actually good for PL Jews, 1450–1772 ?

- Occupational pattern (Roy model)

from moneylenders to leaseholders: Jews became specialized in leasing and managing the nobles' monopolies

- Residential pattern (labor mkts equilibrium)

the rise of myriad shtetls over the vast territory from Baltic to Black Sea

- Population dynamics

Jews: from 0.13% of tot pop to 5.6% of total pop

7. On the road to modernity, 1772 - 1850

- Partitions brought disintegration of the former PLC
- Republic of nobles and economics of nobility gone

➢Occupational pattern (Roy model, again)

from leaseholding to commerce, crafts, high-professionals:

➢Residential pattern (labor mkts equilibrium)

Shtetl still main feature but also migrations to large cities (Warsaw, Odessa, etc.)

➢Population dynamics

Jews: from 5.6% of tot pop to 13% of total pop

Epilogue: From shtetl to New York and Tel Aviv, 1850 – today: Open Questions

Puzzle 1: 1492 - 1850

Similar religious norms for infant/child care and children education (= culture) across world Jewry.

Why only in PL this exceptional demographic growth?

Hp: (external) institutional environment matters!

Puzzle 2: 1850 – today 2 major migrations

Today about 46% of Jews live in Israel

Today about 46% of Jews live in the U.S.

16 million Jews, the start-up country, etc.

Take-home message 1

NATURE or NURTURE?

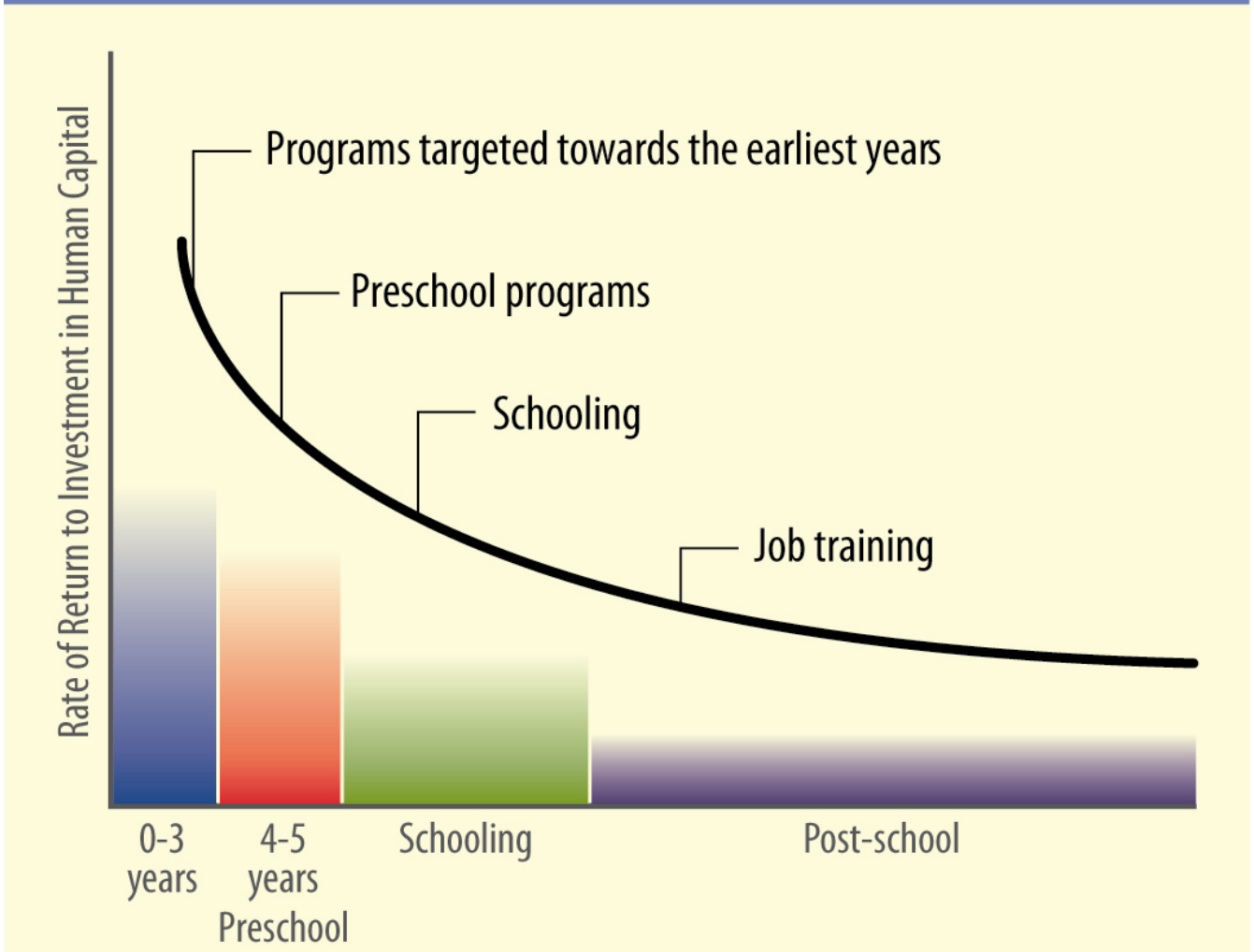
Nurture

- parental investment in
  - infant and child care
  - children education

  fostered by religious / social norms

…. lessons for today?

RATES OF RETURN TO HUMAN CAPITAL INVESTMENT AT DIFFERENT AGES



SOURCE: J.J. Heckman, "Skill formation and the economics of investing in disadvantaged children." *Science*, 312(5782):1900-2, (June 2006).

Take-home message 2

CULTURE or INSTITUTIONS ? Both

Jewish parental investment in

- infant and child care
- children education

fostered by religious / social norms (culture)

&

- Jewish communal (internal) institutions
- economic and political institutions of Muslim Empire and Polish-Lithuanian Commonwealth

Wishful thinking….
The 12-year cycle…



OK, you convinced us!!!
But when will *The Chosen Many* be in the bookstores?





Soon!!!!!

